Multilevel leadership and organizational effectiveness in Indian technical education: the mediating role of communication, power and culture

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ABSTRACT

This study provides a new conceptualization of educational leadership with a multilevel and integrative approach. It examines the impact of multilevel leadership (MLL) on the effectiveness of technical educational institutes through the mediating effects of organizational communication, bases of power and organizational culture. Data were collected from 153 heads of engineering and management institutes across India along with 306 subordinates working in coordination with the heads of the institutes. The subordinates assessed leadership and organizational communication and the leader assessed the remaining constructs. An inventory is developed and validated to assess MLL. The MLL directly predicts organizational effectiveness (OE). Organizational communication, use of personal power and organizational culture mediate the effects of leadership on OE. Multilevel leaders can promote organizational communication, use of personal power and organizational culture for higher effectiveness of technical educational institutions.

Introduction

Without leadership, an organization can hardly succeed because there would be no direction, policies, standards or processes for others to follow and achieve organizational goals. Higher educational institutes like engineering and management schools have the features of both manufacturing and service organizations while being non-profit in nature. These institutes render their service to society by creating and disseminating knowledge through teaching, training and research. They also produce products in terms of human resources, new knowledge and technology. In contemporary situations, universities or colleges operate like corporations, dealing with activities from planning, creating strategies, recruiting and advertising, to managing revenues, faculty, staff and students. In order to promote an effective higher educational institution, a leader must have the skills, knowledge and leadership competencies relevant to the context of higher education (Bryman & Lilley, 2009). Accordingly, the conceptualization of leadership needs to be revisited and new dimensions be added to ensure the effectiveness of technical educational institutions.

The higher technical institutes in India, which include engineering, management, computer applications, pharmacy, architecture and hotel management, have grown sharply in recent years, from 4491 institutions in 2006–2007 with an intake capacity of .75 million to 8562 institutions in 2012–2013 with

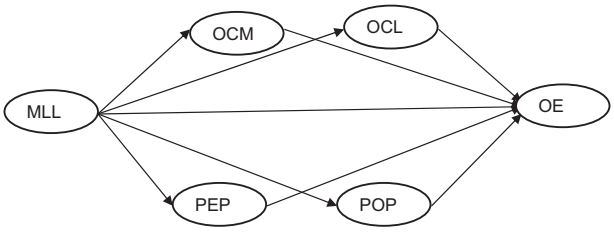

Figure 1. Conceptual model for investigation.
Note: OCM = Organizational communication; PEP = Personal power, POP = Positional power; OCL = Organizational culture

an intake capacity of 2.23 million (AICTE, 2013–14). Higher technical education is passing through a transitional phase as new private universities have been providing such education. The entry of new institutions raises concerns about the quality of education, the management of human, physical and financial resources; and turning students into competent citizens. It has also been emphasized for quality administrators with a strategy and vision for the smooth functioning of higher educational institutions in India (Banerjee & Muley, 2008). These institutes have staffs and students from different parts of the country with diverse cultures, ethnicities and value systems. Managing a diverse population needs special skills, knowledge and attitudes of the administrators/leaders.

First, leaders at the top in technical institutions particularly the directors, principals and deans, involve in faculty and staff recruitment and infrastructure development, address stakeholders' claims, execute strategic decisions, and implement structural changes, set up performance goals, standards and procedures, coordinate organizational processes, and monitor employees' compliance with rules and norms beyond teaching, research, training and outreach programmes. However, most theories of leadership restrict to one or two contrasting attributes (ethical leadership, servant leadership, autocratic vs. democratic, production vs. people oriented, etc.) that partially explain leadership in complex organizational reality. Those put lesser emphasis on situations in which the leader operates. Moreover, cognitive, personality or motivational attributes are found to explain about 30–50% of leaders' performance (Zaccaro, LaPort, & Jose, 2013). To make the leaders more effective, context-relevant political, cultural, technological and financial-performance imperatives can be added (Zaccaro & Klimoski, 2001). A more comprehensive approach is necessary addressing organizational realities and moving beyond the traits and interpersonal attributes of leaders (Bolden & Gosling, 2006).

Second, transformational and charismatic (Antonakis & House, 2002) theories are mostly cited to explain top leadership. Transformational and charismatic leaders have similar attributes such as vision, role modelling, intellectual stimulation, inspiration and setting high expectations (Conger, 1999; Rowold & Heinitz, 2007), and have a similar kind of influencing process that empowers rather than controls subordinates. Even the dimensions of servant leadership focus on individual growth and out-of-box thinking which overlap with transformational leadership's individualized consideration and intellectual stimulation, respectively (Smith, Montagno, & Kuzmenko, 2004). Transformational leadership is found to be more effective than charismatic and transactional leadership in predicting organizational performance (Rowold & Heinitz, 2007). Managerial practices survey, adding more dimensions to transformational leadership, better predict managerial effectiveness than the transformational leadership (Michel, Lyons, & Cho, 2011). However, managerial practices survey dimensions, such as short-term planning, clarifying roles, consulting and external monitoring, are not so relevant to the principals and deans of Indian technical institutions. Such evidences suggest a re-conceptualization of leadership (Rafferty & Griffin, 2004) for organizational effectiveness (OE).

The taxonomy of leadership theories is categorized on the basis of traits, behaviours, power influence and situations (Yukl, 2011). Leadership theories like authentic, ethical and level 5 emphasize the traits of a leader to be successful. Theories like transformational, leader–member exchange, nurturant-task, participative and servant leadership emphasize the behaviour of leaders with their subordinates. Transactional and autocratic leadership have a power and influence approach, whereas situational

leadership portrays a leader's effectiveness as contingent on tasks, situations and cultures. A leadership theory that includes all of these approaches could be more effective, but is still rare to find (Yukl, 2011). Moreover, leadership theories are classified using an individual process, a dyadic process or an organizational process (Yukl, 2011). While an individual process of leadership describes the leader's traits, skills, motives and decision styles, a dyadic/interpersonal process focuses on relationships between the leader and the led. In most theories, an organizational perspective of the leader appears to be missing. A multilevel leadership (MLL) model can include constructs from all levels for higher order effectiveness (Klein, Dansereau, & Hall, 1994; Yukl, 2011). This study intends to develop a MLL model by expanding the full-range model of transformational leadership. To address these concerns, emerging attributes have been discussed for their inclusion in MLL.

Goleman (2011) has emphasized that a leader needs to have emotional intelligence to be effective. It can help leaders solve complex problems, make better decisions, plan how to use time effectively and manage crises (Yukl, 2008). People with high emotional intelligence recognize their own and others' feelings, emotions and drives, and get to know how these factors affect them as well as others. They accordingly mould their behaviour to suit others and situations and find ways to control and even channel their emotions in useful ways. Moreover, they are able to reduce politics and infighting, create an environment of trust and fairness, and increase productivity (Lane & Klenke, 2004). Employees with high levels of emotional competencies are found to be more effective than those with low levels of competencies (Boyatzis, Goleman, & Rhee, 2000; Prasad, Suar, & Taraban, 2014). But, Van Rooy and Viswesvaran (2004) report a correlation of .19 between emotional intelligence and performance outcomes in a meta-analytic study. However, emotional intelligence enhances transformational leadership that furthers organizational performance (Yitshaki, 2012). As stakeholders in educational settings have different demographics leading to a variation in their emotional needs, expectations and work attitudes (Judge, Scott, & Ilies, 2006; Tan, Foo, Chong, & Ng, 2003; van der Velde, Bossink, & Jansen, 2003), emotional intelligence is essential for the leaders in higher education (Zorn & Boler, 2007) to understand others' emotions and display feelings and behaviours accordingly.

In Indian higher educational institutions, students, teachers and staff are from varied cultures and subcultures that differ in language, ethnicity, religion, political orientation and economic status. In the context of such diversity, cross-cultural sensitivity is another essential trait of directors/principals, and deans of educational institutes . This trait describes a leader's ability to understand and function effectively and appropriately in a wide range of cultures (Zirak & Ahmadian, 2012). People with cultural intelligence can choose accurate behaviours and actions for each specific culture, contributing to the organization's cause.

In the context of technology-driven business, Goldsmith (2010) has identified four things a leader needs to do: (1) understand how the intelligent use of new technology can help the organization; (2) recruit, develop and maintain a network of technically competent people; (3) know how to make and manage investments in new technology; and (4) be a positive role model in leading the use of new technology. The leader's technological orientation tends to influence the led to learn and use the latest technologies to be effective.

In the education industry, students and teachers are the internal customers. The external agencies that recruit students and organizations that use the research outputs are the external customers. It is important that professional educators mould their students as per market demands (Hampton, Wolf, Albinsson, & McQuitty, 2009) and train them to meet the needs of the hour. In today's world, educational institutions can sustain only if they produce knowledge, technology or skilled manpower that can be marketable, and can find their end users. Any product that cannot find its end user may not be worth developing when resources are scarce and limited. Hence, the leader may have the orientation to adjust the educational system, curriculum, training and research to suit the requirements of the external customers, seek feedback and be responsive to situational demands.

The new generation is becoming more reactive to unethical practices at the top. Governments have been voted in and out on the issue of corruption, injustice and unethical practices indicating erosion in moral and ethical values (Sapre & Ranade, 2001). In such a situation, leaders' behaviours

are expected to deliver an impartial and value-based ethical treatment across the workplace (Eberlin & Tatum, 2005). A leader who stands for, fights for and delivers justice will command the same from the followers and exert greater influence over them.

With large number of student population, many institutions have come up in the last decade to accommodate them. Some of the institutes have unqualified teachers, inadequate infrastructure and financial crunch. The technocrats and professionals churned out from these institutes do not have employability skills, contributing to unemployment and pseudo-employment. In this scenario, the head of an institute needs to look at the quality of education, employability of the students and sustenance of the institution in accordance with demands of the market.

With the extended list of characteristics, behaviours and traits—emotional intelligence, cultural intelligence, individualized consideration, intellectual stimulation, idealized behaviour, inspirational motivation, market orientation, technological orientation and justice orientation—we interviewed the principals/deans of 12 engineering colleges in Odisha, India. These characteristics, traits, skills and behaviours were supported by effective administrators in technical education.

The individual-level constructs are emotional intelligence and cultural intelligence. These are the leader's ability to understand others' emotions and behave appropriately with individuals from different cultures and subcultures. Individualized consideration, intellectual stimulation, idealized behaviour and inspirational motivation are interpersonal-level variables, whereas market orientation, technological orientation and justice orientation are organizational-level variables. Similarly, from the point of view of an integrative approach, MLL has trait approach like emotional intelligence; a behaviour approach like individualized consideration, intellectual stimulation, idealized behaviour and inspirational motivation; a situational approach like market orientation, technological orientation and cultural intelligence; and an influence approach like justice orientation (Klein et al., 1994; Yukl, 2011). Incorporating these multiple attributes, this study aimed to develop and validate an instrument to assess MLL.

The effectiveness of technical institutions may be assessed in terms of the achievement of objectives (Bluedorn, 1980), the satisfaction of the stakeholders' goals (Connolly, Conlon, & Deutsch, 1980) and smooth functioning without breakdowns (Nadler & Tushman, 1980). Effectiveness in higher educational institutions depends on myriad factors, like the utilization of resources, the quality of the teaching and research, the teachers' competency and the students' employment (Uline, Miller, & Tschannen-Moran, 1998). Furthermore, the leader's way of functioning makes a difference in OE (Thomas, 1988). A transformational leader inspires followers and takes them to a higher plane, and accordingly, followers release effort and energy to realize the organization's goals. Positive leaders' behaviours may increase subordinates' self-esteem, morale and drive for goal accomplishment, reduce absenteeism, create a more cohesive work environment, increase trust between followers and leaders, and increase the followers' feelings of empowerment (Butler, 2009). MLL, which incorporates context-relevant attributes, is likely to predict OE directly. Hence,

H_1: MLL will directly predict OE.

The relationship between MLL and OE is quite intriguing. Organizational communication, bases of power and organizational culture may have roles in the relationship between MLL and OE.

Organizational communication, including managerial, interpersonal and other forms of communication, carries policies, strategies, instructions and information across the organization. It is necessary for the leader to facilitate two-way communication and specifically allow feedback in the form of ideas, suggestions and even dissent. Leaders who communicate frequently help employees to understand what the leaders want, what the organization needs and how the employees fit into the big picture (Baldoni, 2002). As a leader generates thoughts and ideas, these are communicated to stakeholders and attract similar thoughts and ideas from them. Even high-performing, multinational enterprises are characterized by leaders who are good communicators (Rodsutti & Swierczek, 2002). The leader's inspiration, direction, job allotment to and relationship with employees pass through communication to mobilize followers. While effective communication strengthens the leader–member relationship, ineffective communication may result in poor relationships, a possible loss of performance and hamper OE (Ahmed, 2010). Accordingly,

H_2: Organizational communication will mediate the relationship of MLL with OE.

While leadership is a reciprocal influencing process, power is the leader's ability to exert the influence. The bases of power can be deconstructed into positional (legitimate, reward, coercive power) and personal power (expert, referent power). It is more likely that a transactional leader may have a preference for reward and coercive power, a bureaucratic leader may use legitimate power more than other power and a charismatic/transformational leader may prefer expert and referent power more. Organizational perspectives of the multilevel leader may influence the subordinates to consider the leader a role model, bestowing more personal power on him or her. The use of expert, referent and reward power is likely to be beneficial for both personal and OE, whereas the use of coercive and legitimate power may not mobilize the subordinates for OE (Jain, Giga, & Cooper, 2011). The use of expert knowledge and information has the most positive effect on the subordinates' cognitive processes and innovative behaviours that contribute to OE (Krause, 2004). Accordingly,

H_3: Bases of power will mediate the relationship of MLL with OE.

Organizational culture, which is comprised of the common values, beliefs, attitudes, artefacts, customs and practices in an organization, can influence the leader and vice versa. An effective leader can control, reshape or even build a new culture (Bass & Avolio, 1993). Oppositely, a strong culture that is deeply rooted in an organization can change the functioning of a leader. Because higher technical education is undergoing a transformation in India, a leader can reinforce norms and influence employees' behaviours to reshape the culture of the institution. A leader may exert influence on cultural dimensions in the organization, from the vision, mission, adoptability and consistency to the discipline and functionality of the organization. Multilevel leaders may seek a strong organizational culture to promote and pursue their ideas. Attributes like market orientation and technological orientation can nurture a culture that will facilitate the quick adoption to new technologies, processes and/or other changes. This is the reason why high-performing multinational enterprises emphasize rebuilding the organizational culture to be effective (Rodsutti & Swierczek, 2002). An organization's adaptability and consistency influence its ability to develop successful and innovative products. Similarly, a cultural orientation that values team orientation, empowerment and capability development helps in developing positive employee attitudes (Yilmaz & Ergun, 2008). A positive change in organizational culture can set the momentum for increased performance. Given the understanding that a leader influences OE via organizational culture, we propose:

H_4: Organizational culture will mediate the relationship of MLL with OE.

To sum up, this study intends to develop and validate an instrument to assess MLL. In contrast to the direct relationship between MLL and OE, it pleads for an indirect relationship of MLL via communication, power and culture to predict OE. The conceptual model is given in Figure 1.

Methods

Sample

Data were collected from the heads of selected technical and management educational institutions (directors, principals, deans) in India and two of their subordinates (department heads, professors) through a questionnaire survey. The data were collected in person and online using a Google form. Of the 1220 higher technical institutes in Odisha, West Bengal, Uttar Pradesh and Delhi, the researcher approached the heads of 188 randomly selected institutions in Bhubaneswar, Odisha; in Kolkata, West Bengal; in Greater Noida, Uttar Pradesh; and in Delhi, and 535 subordinates therein who were professors or associate/assistant professors. They were briefed about the purpose of the study, assured anonymity of their responses, handed over the questionnaire and asked to return the completed questionnaire in a fortnight. The heads of 47 colleges and 128 professors or associate/assistant professors returned completed questionnaires after repeated requests.

Because of inadequate responses, emails were sent to the heads and subordinates of engineering and management institutes selected randomly from 4160 institutes in the states of Andhra Pradesh, Tamil Nadu, Karnataka, Kerala, Uttar Pradesh and Madhya Pradesh in India. With a cover letter, 633 heads and 1899 professors and associate/assistant professors of the same institutes were approached to complete the questionnaire in online form. Only 121 leader responses and 324 subordinate responses were received.

Of the168 (121 + 47) responses from the heads of the institutes and 452 (324 + 128) responses from the subordinates, the completed questionnaires of 153 heads and their 306 subordinates were considered. An institution was the unit of the sample.

Of the respondents, 143 (93.46%) leaders and 244 (79.73%) subordinates were men. The leaders' average age was 52.66 years and the subordinates' average was 41.11 years. For the leaders, the average professional experience in teaching and years of education in formal educational institutions were 27.66 and 19.36 years, respectively. For subordinates, the average professional experience in teaching and years of education were 17.46 and 18.32 years, respectively.

Measures

With socio-demographics, the survey instrument for the leader contained inventories to assess organizational culture, bases of power used by the leader and OE. The survey instrument for two subordinates who worked under the leader in each institution was used to assess MLL of the superior (director, principal, dean) and organizational communication. The response descriptions for items were on a 5-point Likert scale (1 = Strongly disagree/Very low/Highly dissatisfied, 5 = Strongly agree/Very High/Highly satisfied). Index of inter-rater agreement (r_{wg}) was calculated for MLL and organizational communication items. The median r_{wg} values ranged from .84 to .91 for MLL dimensions and from .85 to .89 for organizational communication dimensions, which indicated good agreement between raters. ICC (1), a form of inter-rater reliability, ranged from .87 to .91 for both MLL and organizational communication dimensions suggesting good inter-rater reliability. Accordingly, the responses of two subordinates of a leader from each institute were averaged to form a single response to be analysed along with the responses from the leader.

To measure *MLL*, 40 items (see Table 2) were developed on nine dimensions. The *communication satisfaction questionnaire* (Downs & Hazen, 1977) was used to measure the organization's communication process. It contained 32 items on eight dimensions. A sample item includes, 'Extent to which conflicts are handled appropriately through proper communication channels'. Hinkin and Schriesheim's (1989) questionnaire based on French and Raven's (1959) conceptualization was used to assess the *bases of power* of the top leadership. It contained 20 items on five bases of power. A sample item includes, 'I sometimes provide my subordinates with sound job related advice'. *Organizational culture* was adapted from Dension's organizational culture survey (Dension & Neale, 1999). To suit the Indian context, the culture survey questionnaire was given to three experts. Thirty-six items out of 60 items agreed upon by the experts were retained on four dimensions and three items were developed according to the experts' advice to assess discipline. A sample item includes, 'There is a clear and consistent set of values that governs the way we run our organization'. Mott's effectiveness questionnaire (1972) was adapted to measure *OE*. It was modified before use to suit the educational sector. Two more items were added on the quality of teaching and research. A sample item includes, 'Do the people in your institution seem to get maximum output from the resources (people, money, equipment, etc.) available? That is, how efficiently do they do their work?'

Results

Partial least squares (PLS) structural equation modelling was used to examine the research model. PLS accommodates both formatively and reflectively measured constructs and has a lower restriction regarding sample size. Also, it does not assume normal distribution of data (Chin, Marcolin, &

Newsted, 2003). SmartPLS 2.0 software was used. The hierarchical component model and a two-stage approach to it were used (Ringle, Sarstedt, & Straub, 2012). In the first stage, the reliability and validity of the first-order reflective indicators were established. In the second stage, the latent scores of the first-order variables were used as formative indicators for second-order constructs.

Measurement model: first-order variables

All the first-order variables were measured reflectively. For each measurement scale, a confirmatory factor analysis was carried out to assess indicator reliability as well as convergent and discriminant validity.

Of the 40 items of MLL, only 16 items had outer loadings above .7. Of the remaining 24 items, 15 items had outer loadings above .4 and the deletion of these items decreased the composite reliability of respective dimensions, hence they were retained. The remaining nine items were dropped for their low outer loadings (<.7). The MLL construct had 31 items, 9 dimensions, and each dimension had at least 3 indicators/items (see Table 1).

All the variables of the study had a composite reliability above .7, confirming internal consistency (Table 2). For all variables, items with outer loadings below .7 were dropped, except for a few items with outer loadings above .4. The deletion of these items caused a decrease in the composite reliability, hence those were retained. The average variance extracted (AVE) of all variables except for five first-order variables were above .5 (Table 2). Those five variables had an AVE close to .5 ensuring their convergent validity and a composite reliability well above .7. The discriminant validity of each variable was ensured because indicators loaded much higher on the associated variable than those loaded on other variables (their own loadings were higher than their cross-loadings) (Chin, 2010). The square root of the AVE of a variable exceeded the inter-correlations with other variables, suggesting the discriminant validity of the first-order constructs.

Measurement model: second-order construct

In the higher-order model, the latent scores of the first-order variables were used as indicators of the second-order construct (Henseler & Chin, 2010). In the higher-order model, all the constructs used formative indicators because the items captured different themes of each construct and were irreplaceable (Hair, Hult, Ringle, & Sarstedt, 2014). Furthermore, some dimensions were not correlated; some were negatively correlated, suggesting their formative approach. As the measures of indicator reliability and discriminant validity were not applicable to formative indicators, we checked for convergent validity, collinearity, relevance and significance. A redundancy analysis was done to test the convergent validity of all the formatively measured constructs. The coefficients of the construct ranged between .99 and 1.0 indicating their convergent validity. All the constructs were tested for collinearity using SPSS 16 and all the variance inflation factors were found to be below 3, rejecting the possibility of collinearity problems and accepting singularity of and relatedness among variables. All the indicators had significant outer weights, suggesting their relative importance. The rest of the indicators with insignificant outer weights had significant outer loadings (p = <.05), indicating their absolute importance (Hair et al., 2014). Thus, all indicators and constructs had acceptable reliability, validity, and were retained.

Data analysis and hypotheses testing

The Pearson correlations revealed that all the nine dimensions of MLL were correlated positively with one another. The emotional and cultural intelligence dimensions were more potently associated with the four dimensions of transformational leadership—individualized consideration, intellectual stimulation, idealized behaviour and influence, inspirational motivation—than with the remaining five

Table 1. Outer loadings of MLL items.

Dimension	Items	Outer loadings	Retained/ Dropped
	The Director/Principal/Dean of the institution …		
Emotional intelligence	… always recognizes his/her emotions and feelings	.30	Dropped
	… can very well understand the feelings and emotions of others	.69	Retained
	… can control his/her emotions and channel them in fruitful ways	.39	Dropped
	… always keep motivating him/herself to improve further	.65	Retained
	… tries to understand others putting him/her in their positions	.79	Retained
	… likes to establish relationships with genuine persons	.65	Retained
Individualized consideration	… doesn't like to delegate any of his/her responsibilities to subordinates even if they are capable of doing it (–)	.16	Dropped
	… stands by every individual and guide them when they need him/her	.80	Retained
	… is involved in mentoring programmes to mentor the new entrants	.59	Retained
	… acknowledges individual's performance in team or organization	.81	Retained
Intellectual stimulation	… advises all to be aware of their own strengths and weaknesses	.59	Retained
	… encourages looking at the problems from different perspective	.75	Retained
	… accepts his/her mistakes and tries to learn from others	.75	Retained
	… always prefers the established and tested ways in comparison to new and risky (–)	.27	Dropped
Idealized behaviour	… sets examples in words and deeds for others to emulate	.58	Dropped
	… as a colleague, is a source of pride for others	.53	Dropped
	… always sticks to his/her words and values irrespective of the situations and consequences	.67	Retained
	… values his/her integrity more than his/her position in the organization	.79	Retained
	… Preaches and practises humility	.62	Retained
Inspirational motivation	… encourages a vision which is in the best interest of the organization	.80	Retained
	… makes all enthusiastic about the vision of the organization	.52	Dropped
	… believes the vision should be clearly understandable, acceptable, and achievable	.72	Retained
	… elevates colleagues/subordinates to a higher level through inspirational dialogues with them	.61	Retained
	… is forward looking for this organization to move ahead	.80	Retained
Justice orientation	… sometimes gives some extra advantage to his/her in-group members than others as they are close to him/her and carry out his/her every wish (–)	.46	Retained
	… does not press hard to do the right thing when it is not possible to do it (–)	.54	Retained
	… thinks the fairness of processes and practices may sometimes be compromised to punish or reward the employees (–)	.57	Retained
	… believes individuals can be treated differently during implementation of any procedure (–)	.47	Retained
	… is impartial in his/her decisions and behaviour	.80	Retained
Market orientation	… emphasizes on research for new technology, process, and/or knowledge to meet the requirements of industries and equip the students with that knowledge	.58	Retained
	… feels that needs and requirements of the recruiters and society are the most valuable inputs for restructuring our course curriculum	.79	Retained
	… acts promptly against complaints and criticisms about our institution	.54	Dropped
	… tries to get feedback from the students as well as experts to improve upon	.74	Retained
Technological orientation	… inspires colleagues and staff continuously to manage technology strategically to achieve our goals	.74	Retained
	… prefers technologically oriented people than others while recruiting	.49	Dropped
	… is a role model in leading the use of new technology in our institution	.61	Retained
	… feels that it is not necessary for him/her to be technology savvy (–)	.72	Retained
Cultural intelligence	… perceives it natural, if he/she encounters something unexpected while dealing with people from different cultures/subcultures	.81	Retained
	… feels free and confident to deal with persons from different cultures/subcultures	.71	Retained
	It is difficult for him to change his body language, signs and/or symbols to suit people of different cultures/subcultures (–)	.70	Retained

(–) Indicates a negatively keyed item.

Table 2. Psychometric properties for first-order construct.

Constructs	AVE	Composite reliability	*M*	SD
Multilevel leadership				
Emotional intelligence	.50	.80	4.11	.57
Individualized consideration	.55	.78	4.05	.58
Intellectual stimulation	.50	.75	4.11	.58
Idealized behaviour and influence	.56	.79	3.96	.67
Inspirational motivation	.56	.83	4.07	.60
Justice orientation	.35	.72	3.17	.68
Market orientation	.58	.81	4.11	.64
Technological orientation	.50	.79	3.55	.57
Cultural intelligence	.56	.79	3.45	.64
Organizational communication				
Organizational integration	.54	.78	4.01	.67
Organizational perspective	.62	.83	3.92	.69
Personal feedback	.63	.83	3.83	.77
Media quality	.50	.75	3.86	.60
Horizontal and informal communication	.52	.76	3.84	.59
Communication climate	.57	.80	3.88	.64
Relationship with subordinates	.48	.79	3.87	.61
Relationship with superiors	.55	.78	3.97	.61
Organizational culture				
Discipline	.61	.82	3.81	.72
Adaptability	.47	.84	3.93	.51
Involvement	.63	.90	4.06	.64
Mission	.54	.87	3.97	.56
Consistency	.45	.80	3.61	.60
Power				
Legitimate	.47	.78	4.20	.48
Reward	.57	.80	3.39	.63
Coercive	.66	.89	2.03	.81
Expert	.62	.86	4.05	.52
Referent	.63	.87	4.07	.63

Table 3. Correlation among MLL dimensions.

Dimension	1	2	3	4	5	6	7	8	9
1. Emotional intelligence	1	.63***	.62***	.56***	.49***	.15	.39***	.26***	.23**
2. Individualized consideration		1	.56***	.52***	.52***	.23**	.41***	.38***	.33***
3. Intellectual stimulation			1	.49***	.39***	.18*	.46***	.17*	.31***
4. Idealized behaviour and influence				1	.51***	.11	.38***	.31***	.26***
5. Inspirational motivation					1	.13	.65***	.35***	.24**
6. Justice orientation						1	.20*	.33***	.26**
7. Market orientation							1	.35***	.17*
8. Technological orientation								1	.03
9. Cultural Intelligence									1
M	4.11	4.05	4.11	3.96	4.07	3.17	4.11	3.55	3.45
SD	.57	.58	.58	.67	.60	.68	.64	.57	.64

*$p = <.05$; **$p = <.01$; ***$p = <.001$.

dimensions in MLL. Only emotional intelligence as well as idealized behaviour and influence were not associated with justice orientation and so also technical orientation with cultural intelligence (Table 3).

When the newly proposed five dimensions in MLL entered stepwise followed by four dimensions of transformational leadership, the five dimensions explained 42.8% variance of OE. Conversely, when the four dimensions of transformational leadership entered stepwise followed by newly proposed five dimensions in MLL, the four dimensions explained 31.6% variance of OE. All the nine dimensions of MLL explained 45.7% variance of OE. First, such observations suggested that the newly proposed five dimensions in MLL explain more variance of OE than the four dimensions of transformational

Table 4. Dimensions of MLL construct predicting organizational effectiveness.

Step	MLL dimensions	R^2	ΔR^2	Step	MLL dimensions	R^2	ΔR^2
1	Emotional intelligence	.249	–	1	Idealized behaviour and influence	.191	–
2	Cultural intelligence	.331	.082	2	Inspirational motivation	.291	.10
3	Technological orientation	.366	.035	3	Individualized consideration	.294	.003
4	Market orientation	.421	.055	4	Intellectual stimulation	.316	.022
5	Justice orientation and execution	.428	.007	5	Emotional intelligence	.335	.019
6	Idealized behaviour and influence	.428	0	6	Cultural intelligence	.401	.066
7	Inspirational motivation	.437	.009	7	Technological orientation	.426	.025
8	Individualized consideration	.457	.020	8	Market orientation	.450	.024
9	Intellectual stimulation	.457	0	9	Justice orientation and execution	.457	.007

Table 5. Mediating effects of communication, power and culture between MLL and OE.

Step	Path	Path coefficient	*t*-statistics	Mediating effects (%)	Inference
1	MLL→OE	.68	17.67***		H_1 Supported
2	MLL→OCM	.84	28.88***		
	MLL→PEP	.61	10.92***		
	MLL→POP	−.55	9.27***		
	MLL→OCL	.61	13.58***		
3	OCM→OE	.35	4.61***		
	PEP→OE	.30	2.09*		
	POP→OE	−.07	.96		
	OCL→OE	.25	2.51*		
4	MLL→OE	−.05	.51		
	MLL→OCM	.83	26.43***	45.22	H_2 Supported
	OCM→OE	.36	3.20***		
	MLL→PEP	.61	10.76***	34.18	H_3Partly supported
	PEP→OE	.36	2.25*		
	MLL→POP	−.52	8.16***		
	POP→OE	.03	.34		
	MLL→OCL	.60	13.11***	22.54	H_4 Supported
	OCL→OE	.25	2.38*		

Note: OCM = Organizational communication, PEP = Personal power, POP = Positional power, OCL = Organizational culture.
*p = <.05; **p = <.01; ***p = <.001.

leadership in MLL (Table 4). Moreover, when the newly proposed five dimensions and the four dimensions of transformational leadership in MLL were used as two latent constructs to predict OE, the former directly predicted OE ($\beta = .58$, $p < .001$), but the latter did not ($\beta = .09$, $p = .28$). The effect-size (Cohen, 1988) for the former was medium ($f^2 = .23$) and the latter was negligible ($f^2 = .01$), suggesting the higher predictive relevance of newly proposed five dimensions than the four dimensions of transformational leadership in MLL. The newly proposed five dimensions absorbed the variances of four dimensions of transformational leadership in MLL.

To examine the mediating effects of organizational culture, organizational communication and bases of power between MLL and OE, Baron and Kenny (1986) arguments were followed. First, the independent variable of MLL would predict the dependent variable of OE. Second, the independent variable would predict the mediating variables of organizational communication, personal and positional power and organizational culture. Third, the mediators would predict the dependent variable. In the fourth step, when the independent variable and mediators would jointly explain the OE, the effects of independent variable explaining OE will decrease (partial mediation) or nullify (full mediation). The extent of mediation was estimated by multiplying the indirect paths between MLL and OE and dividing that by the total effect and multiplied by 100.

In accordance with the first hypothesis and the first step of mediator analysis, MLL was positively associated with OE. In accordance with the second step of mediator analysis, MLL positively predicted organizational communication, use of personal power and organizational culture, but negatively predicted the use of positional power. In accordance with the above-stated third step, better organizational

communication, more use of personal power and stronger organizational culture were positively associated with higher OE. Positional power, however, did not predict OE (Table 5).

In the full model (Figure 1), with organizational communication, bases of power, organizational culture and MLL as explanatory variables, MLL did not predict OE and the sign of the standardized beta changed. Therefore, in accordance with the last three hypotheses, organizational communication, organizational culture and personal power fully suppressed the effects of MLL on OE. The third hypothesis was partially supported because personal power mediated the relationship of MLL with OE, but positional power did not. The mediating effects of organizational communication, personal power and organizational culture were found to be 45.22, 34.18 and 22.54%, respectively, indicating that organizational communication was the strongest mediator of the relationship of MLL with OE. The predictive accuracy of the full model was substantial (R^2 = .60).

Discussion

Studying 153 heads of higher technical institutions and their 306 subordinates in India, this study develops and validates a scale for assessing MLL. The findings suggest that MLL better predicts directly OE than its transformational attributes. The effect of MLL construct passes through organizational communication, organizational culture and the use of personal power, but not through the use of positional power in furthering the effectiveness of higher educational institutions.

Though transformational and charismatic theories (Antonakis & House, 2002) are often found to be effective for the leadership at the top (Michel et al., 2011), these theories do not encapsulate the contemporary realities in higher educational institutions. MLL addresses this concern with the inclusion of market orientation, justice orientation and technological orientation. The intra-individual attributes of emotional and cultural intelligence are added to the interpersonal aspects. Furthermore, MLL integrates trait, behaviour, situational and influence approaches as well as personal, interpersonal and enterprise-level constructs.

When four dimensions of transformational leadership and rest five dimensions of MLL were considered jointly, transformational leadership dimensions did not predict OE, suggesting that the style alone is not sufficient to make educational institutions effective. Attributes added to transformational leadership make it stronger when predicting OE. Having emotional intelligence, the leader can understand and appreciate the emotions, attitudes and behaviours of self as well as others and display the context-relevant emotions, attitudes and behaviours for higher effectiveness. A market orientation makes leaders more need focused, and keeps them updated, helping the educational institution to be competitive. A leader, who has a concern for the needs of the society, seeks feedback, accepts criticism and suggestions, and incorporates changes, leads to higher institutional effectiveness. Similarly, a justice orientation adds strength to the leadership to act unbiased against all odds. A leader with a technological orientation may have the advantage over leaders who believe in conventional skills and may influence the organization to strive for more advanced and effective technological adoption, leading to OE. Leaders' emotional intelligence and cultural intelligence help them to display more appropriate behaviours in interpersonal context.

Evidence suggests that leadership effectiveness is positively associated with communication, personal power and culture (Boyatzis, 1971). The last three constructs are found to influence positively the OE (Ahmed, 2010; Jain et al., 2011; Rodsutti & Swierczek, 2002; Yilmaz & Ergun, 2008). The findings here suggest that the communication, personal power and culture are the vehicles through which the effects MLL pass on to OE.

A better communication process enables the leader to send the message precisely, helping the leader to effectively use his or her skills, traits and attributes for higher OE, whereas poor communication may make a better leader worse, deteriorating his/her sphere of influence and leading to poor OE. A case in point is the collapse of Enron. Enron collapsed as senior managers did not communicate organizational problems to employees for their resolution (Seeger & Ulmer, 2003). Also, Piyasvasti Amranand lost his job as CEO of Thai Air, in May 2012, even after getting a positive annual review

by the company's board. The chairman of the airline reported to the media that communication problems between Piyasvasti and the board were hampering the company's performance that led to his dismissal. With open communication and coordination (Flin & Yule, 2004) between the leader and the led, MLL furthered most potently the OE.

Subordinates are more influenced and inspired by acts of expertise and appealing personal qualities than by orders, punishments and transactional actions. With individual and interpersonal attributes, a multilevel leader is more likely to use his or her personal power contributing to OE. Dr A P J Abdul Kalam, a recipient of *Bharat Ratna*, highest civilian award of India, is a distinguished researcher and educational leader. During his tenure as director and chief project coordinator at Defence Research and Development Organisation, he hardly used his positional authority; rather, his expertise and personal charm inspired the subordinates to work for the success of projects. Even after his retirement from the position, people take pride to have worked with him and the incidences of his personal influence are told to organizational members again and again. In line with such an observation, use of personal power strongly bridges the relationship between leadership and OE.

Culture is created by shared experience, but it is the leader who initiates the process by inculcating his/her beliefs, values, assumptions and learning experiences (Schein, 2004) in employees. Cultural norms arise and change according to the leader's concern (Bass & Avolio, 1993). A multilevel leader seeks participation of followers through social and structural bonding. Participation provides a sense of ownership, while a shared sense of purpose, direction and strategy can galvanize the actions of organizational members toward collective goals. A multilevel leader, having an organizational perspective, sets a vision for the organization, prefers a culture with adaptability to meet the needs of the external market, and keeps the organization technologically updated. This in turn contributes to the goal achievement vis-a-vis OE. Ratan Tata, the Chairman Emeritus of Tata Sons, is a standing example of a leader who created a culture in the best interests of the organization. He took over as chairman of the Tata group, in 1991. He inherited many independently run companies, many of which were run by their CEOs under having overlapping businesses and different work cultures across multiple companies. He took bold steps, and sold some businesses, in order to rationalize the functioning of the Tata Group. He emphasized on technology-driven leadership, global competitiveness and employees' welfare. The Tata Group is now more streamlined, having a common Tata culture and identified by Tata brand name. At his retirement on December 28, 2012, the net profit of Tata Group had grown four times the 1991 level.

Implications and limitations

MLL has both an integrative approach and a multilevel orientation along with transformational attributes. People with these attributes are more likely to be effective; hence, they can be preferred for leadership positions to help higher technical institutions become more effective.

This study is not free from limitations. First, the participants of this study were the heads of institutes and the subordinates who work in direct coordination with them. In many private educational institutions, these designated individuals may not have real authority over the management of these institutions. Such cases were hard to identify in this study, and future research may be carried out on this issue. Second, self-reported responses to a questionnaire cannot be free from socially desirable responding. However, the responses to the items on MLL and the communication process were obtained from subordinates and the other constructs from top-level leaders, which procedurally arrested the common method bias. Third, though we assessed the objective indices of OE on infrastructure availability, teacher–student ratio, job placement of students, publications in peer-reviewed journals and handling of projects, the objective indices were not analysed due to availability of data only in 20 organizations. Notwithstanding the limitations, this study developed a context-relevant inventory to assess MLL. The newly proposed five dimensions of MLL are found to be the better predictor of OE than transformational attributes. Moreover, the effects of MLL are found to pass through culture, communication and personal power to help technical educational institutions become more effective.

Disclosure statement

No potential conflict of interest was reported by the authors.

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